DEPARTEMEN TEKNIK ELEKTRO OTOMASI

GUIDEBOOK COMPETITION IARC XII

VORTEX

SMA/K SEDERAJAT
MAHASISWA

iarc.its

BACKGROUND

Current global technological development is in an exponential phase that demands efficiency, precision, and sustainability across various sectors of life. Engineering systems (*engineering systems*) are no longer merely static tools, but have transformed into a key driving force (*catalyst*) capable of transforming civilization and addressing global challenges. In line with this vision, the theme "Engineering System that Moves the World" was adopted as the foundation for innovation in this competition.

In the era of digital transformation, the integration of Artificial Intelligence (AI), robotics, and automation is a key factor in optimizing industrial productivity and solving real problems in society. However, technological innovation is not always about complexity. The development of simple, affordable, and sustainable Appropriate Technology is often the most crucial solution for directly addressing the fundamental needs of society and improving quality of life.

On the other hand, the world today faces real threats from climate change and unpredictable disaster risks. The use of smart systems and automation is absolutely necessary to strengthen mitigation, preparedness, and rapid response in order to minimize the resulting social and economic impact. Finally, a nation's resilience also greatly depends on the adaptation of responsive defense technology. Through the implementation of robotics and smart systems, future security and defense can be more thoroughly prepared to face the various dynamics of global challenges.

Given the significant urgency across these four sectors, IARC XII Vortex Competition 2026 serves as a strategic platform for innovative young talents at the university level throughout Indonesia. Through the *Vortex Idea Challenge* and Vortex Innovation Challenge, this competition is committed to gathering visionary ideas, sharpening critical reasoning, and testing the feasibility of cutting-edge technology prototypes capable of driving the progress of the nation and the world.

EVENT NAME

- IARC XII Vortex Innovation Challenge 2026
- IARC XII Vortex Idea Challenge 2026

VORTEX COMPETITION THEME

Main Theme: "*Engineering System that Moves the World*"

Problem Statements

1. **Artificial Intelligence and Digital Technology**

   Problem Statement: The growth of urban areas increases the complexity of managing transportation, energy, the environment, security, public services, and various community activities. The availability of data and the development of digital technology create opportunities to build urban systems that are more integrated, adaptive, and data-driven. Participants are challenged to develop Smart City solutions capable of integrating digital technology, artificial intelligence, the Internet of Things, data analysis, or related technologies to effectively and sustainably address urban problems.

2. **Appropriate Technology**

   Problem Statement: Medical and healthcare problems require not only devices capable of performing basic functions, but also technologies that can provide accurate, safe, and reliable measurement and monitoring results. The integration of sensors, control systems, signal or image processing, and data analysis can be part of the development of healthcare technology to support diagnosis, monitoring, rehabilitation, and the improvement of healthcare service quality. Participants are challenged to develop appropriate technologies capable of integrating these various aspects to address real-world problems in the medical and healthcare fields.

3. **Disaster Mitigation and Climate Change**

   Problem Statement: Earthquake risk is influenced by various factors related to physical conditions, the environment, infrastructure, as well as the characteristics of the region and its communities. This complexity requires an approach that

focuses not only on event detection, but also on risk assessment, preparedness, mitigation, and improving regional resilience. Participants are challenged to develop technology-based solutions capable of supporting earthquake risk reduction through modeling, monitoring, data analysis, vulnerability assessment, warning systems, or decision support systems.

4. **Defense Technology**

Problem Statement: The development of defense technology requires systems capable of enhancing operational capabilities in addressing various conditions and mission requirements. The development of technologies supporting combat capabilities may encompass aspects of systems, sensors, control, communication, materials, as well as the integration of various technologies to enhance operational capabilities. Participants are challenged to develop innovative and integrated technological solutions to support defense system capabilities while considering aspects of effectiveness, reliability, and safety.

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CHAPTER 1 - COMPETITION PROVISIONS

A. Participant Requirements

1. **Academic Status:**
   - Active undergraduate (S1) or Diploma students from any university in Indonesia.
   - Must include proof of active student status in the form of a valid Student ID Card (KTM).
2. **Team Composition:**
   - Participants may take part in the competition individually or in a group (maximum of 3 people).
   - Each team consists of 1–3 members, including one team leader.
   - Each participant may only be a member or leader of one team.
   - Team members may come from different classes or majors, but must remain within the same school.
3. **Participation Requirements:**
   - Participants are not allowed to change team members during the competition.
   - Each team is only allowed to submit a maximum of one entry.
   - Participants must follow the Instagram account @iarc.its and join the official WhatsApp group provided by the committee.
4. **Administrative Requirements:**
   - All participants who advance to the final round must attend the IARC XII 2026 culminating event, as the winners will be announced directly at that event.
   - Participants must upload a Vortex Competition IARC XII 2026 participation twibbon to their personal Instagram account with the hashtag and caption in accordance with the provisions in the guidebook, tag 3 friends, and tag the @iarc.its account.
   - Participants must pay attention to every deadline in the timeline listed in the guidebook (see Chapter 2).
5. **Provisions:**
   - The submitted work must be original, must never have been entered in or won a similar competition, and must not contain elements of SARA (ethnicity, religion, race, or inter-group issues), violence, or pornography.
   - The maximum tolerance for similarity/plagiarism is 20%, as measured using the plagiarism-checking application specified by the committee. Work found to exceed this limit will be disqualified without a refund of the registration fee.

- Participants must fill out and sign the Work Originality Statement Form affixed with a Rp10,000 stamp duty.

6. **Registration Fee**

Category	Competition	Batch	Fee
College	Vortex Innovation Challenge and Vortex Idea Challenge	Batch 1	Rp 75.000,00
		Batch 2	Rp 90.000,00
		Batch 3	Rp 120.000,00

B. **Work Requirements**

1. **Idea Challenge**
    1. **Writing Format**
        - Paper size: A4 (210 × 297 mm) · Margin: 2.5 cm on all sides · Orientation: Portrait
        - Font: Times New Roman, 12 pt · Title: 14 pt Bold · Spacing: 1.15 · Alignment: Justify
    2. **Paper Structure**
        - Cover Page (using the provided template)
        - Abstract (maximum 200 words)
        - Introduction
        - Ideas & Analysis
        - Conclusion
        - References
    3. **File Format**
        - The file is submitted in PDF format
        - File name format: “IARC_[TeamName]_[Category]_Idea_Vortex”
        - Example: “IARC_TeamFuture_HigherEducation_Idea_Vortex”

2. **Innovation Challenge**
    1. **Writing Format**
        - Paper size: A4 (210 × 297 mm) · Margin: 2.5 cm on all sides · Orientation: Portrait
        - Font: Times New Roman, 12 pt · Title: 14 pt Bold · Spacing: 1.15 · Alignment: Justify
    2. **Paper Structure**
        - Cover Page (using the provided template)
        - Abstract (maximum 200 words)
        - Introduction

- Research Method
- Results and Discussion
- Conclusion
- References

3. **File Format**
    - The file is submitted in PDF format
    - File name format:
      "IARC_[TeamName]_[Category]_Innovation_Vortex"
    - Example:
      "IARC_TeamFuture_HigherEducation_Innovation_Vortex"

C. Special Provisions

1. The submitted work is the participant's original work and has never won any competition
2. The work does not contain elements of racism, pornography, or anything offensive to a particular group
3. Teams are prohibited from including institutional identity information within the paper
4. Work proven to be plagiarized will be immediately disqualified (maximum similarity tolerance of 25%)
5. Writing must use proper and correct Indonesian in accordance with PUEBI (the Indonesian spelling guidelines)
6. References must use the APA citation format
7. The jury's decision is final and cannot be contested

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CHAPTER 2 - SCHEDULE

Stage	Date
Open Registration Batch 1	September 25, 2026 - October 16, 2026
Open Registration Batch 2	October 7 2026, - November 7, 2026
Open Registration Batch 3	November 8, 2026 - November 27, 2026
Paper Submission	December 22, 2026
Final Announcement	December 30, 2026
Technical Meeting	December 18, 2026

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CHAPTER 3 - ASSESSMENT CRITERIA

A. Selection Stage Assessment

- **Vortex Idea Challenge**

No	Parameter	Description	Percentage (%)
1	**Writing Format**	Compliance with the specified format, correct spelling, good grammar	10
2	**Writing Systematics**	Systematic writing, clear flow, coherent discussion	15
3	**Relevance & Depth of Issue**	Relevance to the theme; depth of understanding of the issue chosen by the team	20
4	**Originality of Ideas**	Novelty of ideas, creativity of approach, uniqueness compared to existing solutions	35
5	**Argumentation & Concept Feasibility**	Strength of reasoning; plausibility of the idea as a future direction (conceptual in nature, not at the implementation level)	20
Total			**100**

- **Vortex Innovation Challenge**

No	Parameter	Description	Percentage (%)
1	**Writing Format**	Compliance with the specified format, correct spelling, good grammar	10
2	**Research Methodology**	Systematic writing, clear flow, coherent discussion	15

3	**Urgency & Criticality of the Problem**	Relevance to the given case study; sharpness of problem analysis; importance of the solution	30
4	**Solution & Feasibility**	Innovation of the solution, strong argumentation, feasibility of implementation, evidence/prototype	30
5	**Originality**	Novelty of approach, creativity, uniqueness of the solution (maximum plagiarism tolerance of 20%)	15
Total			**100**

B. Final Stage Assessment

- **Vortex Idea Challenge**

No	Parameter	Description	Percentage (%)
1	**Rule Compliance**	Punctuality, presentation format, dress code	5
2	**Spontaneity & Idea Development**	Quality of thinking under time pressure when developing the new case study given in the final round	25
3	**Concept & Solution**	Quality, creativity, relevance, and feasibility of the concept or solution developed to solve the problem.	20
4	**Verbal**	Quality of thinking under time pressure when developing the new case study given in the final round	25
5	**Q&A**	Logical coherence and creativity of the ideas developed	25
Total			**100**

- **Vortex Innovation Challenge**

No	Parameter	Description	Percentage (%)
1	**Rule Compliance**	Punctuality, presentation format, dress code	5
2	**Urgency & Criticality of the Problem**	Understanding of the problem, relevance to the theme	30
3	**Technical**	Quality of the technical solution, feasibility of implementation	15
4	**Verbal**	Communication, delivery of ideas, clarity	25
5	**Q&A**	Ability to answer, depth of understanding	25
Total			**100**

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CHAPTER 4 - COMPETITION STAGES

A. Preliminary Stage

- Participants choose one of the four problem statements provided by the committee, according to the chosen sub-theme.
- Participants develop their work based on the chosen problem statement, in accordance with the established writing provisions.
- The completed work is submitted in PDF format through the official IARC XII 2026 submission form.
- Participants must ensure that the submitted work meets all the format, systematics, and work provisions listed in the guidebook.
- Submitted work will go through an assessment process by the jury to determine which participants are eligible to advance to the final round.
- The 10 best participants in each category, as determined by the committee, will be declared to have advanced to the final round.

B. Final Round

1. Vortex Idea Challenge Final Round

- Participants who advance to the final round must take part in the entire series of final activities in accordance with the schedule and provisions set by the committee.
- In the final round, participants will be given a new case study or problem prepared by the committee.
- Participants are given 120 minutes to understand the problem and develop a solution to resolve the given case study.
- After the working time ends, participants present the ideas or solutions they have developed to the jury using the 3MT (Three Minute Thesis) format.
- Each participant is given a maximum of 3 minutes to present their ideas concisely, clearly, and persuasively.
- After the 3MT session ends, participants take part in a Q&A session with the jury to explain and defend the ideas they have developed.
- The final round assessment is carried out based on the assessment criteria set by the committee.

2. Vortex Innovation Challenge Final Round

- Participants who advance to the final round must present the work or device they have developed in the form of an exhibition booth, in accordance with the theme and provisions set by the committee.

- Participants prepare and set up their booth along with the work/device to be exhibited. Participants are responsible for explaining the concept, background, how it works, advantages, benefits, and the innovation offered by the work or device.
- The jury will go around visiting each participant's *booth* (*booth judging*) to see and assess the work directly.
- When the jury visits the booth, participants will give a live presentation and demonstration of the work or device developed and answer the questions asked by the jury.
- Assessment is carried out directly by the jury based on the criteria set by the committee, covering aspects of
- Each participant must ensure that the work or device can be displayed and/or demonstrated live throughout the judging process.
- The final round assessment is carried out based on the overall judging results of the work/device and the participants' presentation, in accordance with the assessment criteria set by the committee.

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CHAPTER 5 - WINNER PRIZES

Student Category

- 1st Champion: Development Money + E-certificate + Trophy
- 2nd Champion: Development Money + E-certificate + Trophy
- 3rd Champion: Development Money + E-certificate + Trophy

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CHAPTER 6 - CONTACT PERSON

Vortex Innovation & Idea Competition Coordinator

- Syifa
  - Whatsapp : wa.me/6281334937231
- Fikri
  - Whatsapp : wa.me/6281334937231

CHAPTER 7 - REGISTRATION & PAYMENT MECHANISM

1. **Registration Mechanism**
   - Participants register through the official IARC XII 2026 form provided by the committee.
   - Registration is completed by filling in the participant and team member data fully and correctly, in accordance with the provisions stated in the registration form.
   - Participants must complete all registration requirements and upload the required documents through the form.
   - The registration form can be accessed through the following link: **https://its.id/m/FormRegistVortexIARCXII**

2. **Payment Mechanism**

   Payment of the registration fee is made through the official account designated by the committee.

   Bank: BCA
   Account Number: 1430996882
   Account Holder Name: Luna Aulia Afifah

   Registration Fee: [based on page 6]

3. **Payment Confirmation**
   - Participants must upload proof of payment on the registration form.
   - Proof of payment must be clearly legible and show the corresponding transaction information.
   - Payment will be verified by the committee after the participant completes the registration process.
   - Participants who have been verified will receive registration confirmation from the committee.
   - Proof of payment must be kept by the participant as valid evidence of payment.

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CHAPTER 8 - ATTACHMENTS

1. Cover Page

(PAPER TITLE)

Institution Logo
Participant
5x5 cm

IARC Logo
2025
5x5 cm

Proposed By:
Name
NIM/NIS

INSTITUTION NAME
CITY
YEAR

1. Work Originality Statement Sheet

STATEMENT OF WORK ORIGINALITY

The undersigned below:

Full Name (Leader) :

NISN/NIS :

Place/Date of Birth :

School Name :

Email Address :

Phone Number :

Declare that the work entitled "....." is truly our own work, not a copy or plagiarism of someone else's work, either in part or in full, has never won or been entered in a similar competition, and has never been published in any media.

This statement of work originality has been made truthfully. Should it be proven in the future that plagiarism has been committed, we are willing to accept the sanctions set by the committee of the Industrial Automation and Robotic Competition (IARC) XII 2026.

City, date - month - year

The undersigned,

Full Name of Author

1. Biodata of Leader and Members

3.1 Biodata of Leader

1. **Personal Identity**

1	Full Name	
2	Gender	
3	Study Program	
4	Student ID Number (NIM)	
5	Place and Date of Birth	
6	Email Address	
7	Phone/Mobile Number	

1. **Awards Previously Received**

No	Type of Award	Awarding Party	Year

All data I have entered and stated in this biodata form is true and can be legally accounted for. If any discrepancy with the facts is found in the future, I am willing to accept the consequences.

This biodata has been made truthfully to fulfill one of the requirements for the Industrial Automation and Robotic Competition (IARC) XI 2025 in the Essay competition category.

City, Date-Month-Year

Team Leader

Leader's Name

NIM.

3.2 Biodata of Member

A. Personal Identity

1	Full Name	
2	Gender	
3	Study Program	
4	Student ID Number (NIM)	
5	Place and Date of Birth	
6	Email Address	
7	Phone/Mobile Number	

B. Awards Previously Received

No	Type of Award	Awarding Party	Year

All data I have entered and stated in this biodata form is true and can be legally accounted for. If any discrepancy with the facts is found in the future, I am willing to accept the consequences.

This biodata has been made truthfully to fulfill one of the requirements for the Industrial Automation and Robotic Competition (IARC) XI 2025 in the Essay competition category.

City, Date-Month-Year

Team Member

Member's Name

NIM.

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1. Advisor Biodata

1. **Personal Identity**

1	Full Name (with title)	
2	Gender	
3	Study Program	
4	NIP/NIDN	
5	Place and Date of Birth	
6	Email Address	
7	Phone/Mobile Number	

1. **Educational History**

No	Level	Field of Study	Institution	Year Entered – Graduated
1				
2				

2. **Tri Dharma Higher Education Teaching Record**

No	Course/Subject Name	Required/Elective	Credits (SKS)
1.			
2.			

3. **Research**

No	Research Title	Funding Source	Year
1.			

All data I have entered and stated in this biodata form is true and can be legally accounted for. If any discrepancy with the facts is found in the future, I am willing to accept the consequences.

This biodata has been made truthfully to fulfill one of the requirements for the Industrial Automation and Robotic Competition (IARC) XI 2025 in the Essay competition category.

City, date-month-year

Advisor / Lecturer

Name of Advisor / Lecturer

NIP/NIDN.